

8.3.1.1.2.1 Optional Real-Time Calibration Using the Host Microcontroller

The performance of the cell voltage values measured by the 14-bit ADC has a factory-calibrated accuracy, as follows:

- ± 10 mV TYP, ± 40 mV MIN and MAX from 3.6 to 4.3 V,
- ± 15 mV TYP, ± 40 mV MIN and MAX from 3.2 to 4.6 V, and
- ± 50 mV MIN and MAX from 2.0 to 5.0 V

While this is suitable for the majority of pack protection and basic monitoring applications the BQ769x0 AFE family is intended to support, certain systems may require a higher accuracy performance.

To achieve this, use an available ADC channel and general purpose output terminal on the host microcontroller paired with the BQ769x0. A simple external circuit consisting of two precision resistors and a small-signal FET is activated by the host microcontroller to determine the total stack voltage, V_{STACK} . This is then compared against the sum of the individual cell voltages as measured by the internal ADC of the BQ769x0. The resulting transfer function coefficient, $GAIN_2$, is simply applied to each cell voltage ADC value for improved accuracy.

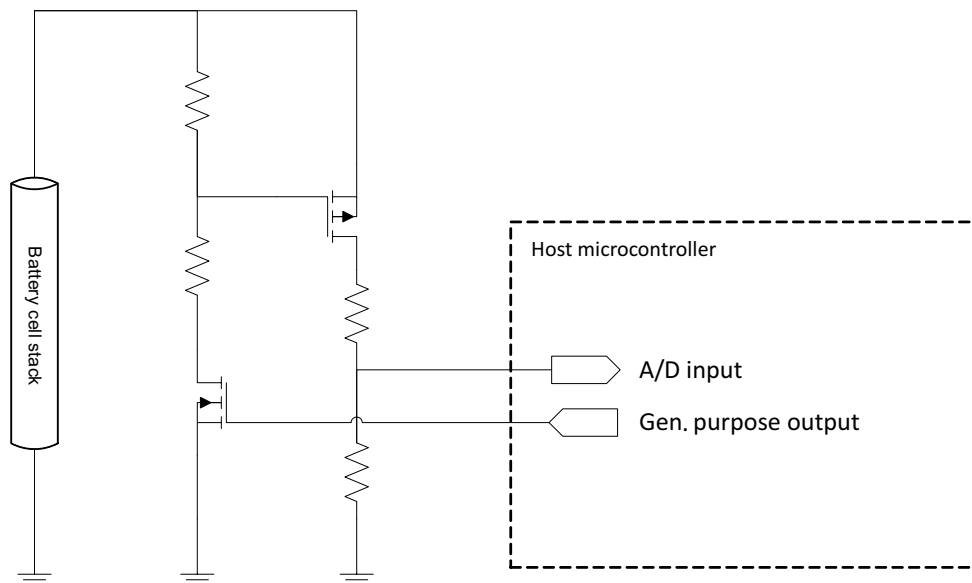


图 8-1. External Real-Time Calibration Circuit to Host Microcontroller

The process is as follows:

1. Periodically measure V_{STACK} .
 - a. $V_{STACK} = V_{AD} \times (R1 + R2) / R1$
2. Read out all V_{CELL} ADC readings from the BQ769x0 and apply the standard GAIN and OFFSET values stored in the BQ769x0.
 - a. $V(1) = GAIN \times ADC_1 + OFFSET$, $V(2) = GAIN \times ADC_2 + OFFSET$, and so on
3. Sum up all V_{CELL} values, V_{SUM} .
 - a. $V_{SUM} = V(1) + V(2) + V(3) \dots$
4. Calculate $GAIN_2$.
 - a. $GAIN_2 = V_{STACK} / V_{SUM}$

As a general recommendation, a new $GAIN_2$ function should be generated when the cell voltages increase or decrease by more than 100 mV. With $GAIN_2$, each cell voltage calculation becomes:

$$V(\text{cell}) = GAIN_2 \times (GAIN \times \text{ADC}(\text{cell}) + \text{OFFSET}) \quad (2)$$

For systems that do not require this additional in-use calibration function, $GAIN_2$ is simply “1”.